

Research Statement

Dr. rer. nat. Martin Bies

Last updated: September 13, 2024

Since my undergraduate studies, *string theory* has fascinated me as a potential unified theory of nature. A key aspect of string theory is its vast array of solutions, known as the *string landscape*. My research focuses on identifying string theory solutions consistent with observed physics or proving their absence. To achieve this, I rely on geometric methods, particularly toric and algebraic geometry – from both a practical and a theoretical perspective. I specialize in constructive and enumerative approaches, using computational tools to identify optimal string theory solutions.

My research spans three areas: physics, focusing on *F-theory* (a set of string theory solutions encoded in elliptic fibrations); mathematics, with emphasis on toric and algebraic geometry, and connections to combinatorics, graph theory, and number theory; and computer science, particularly open-source computer algebra systems (most recently OSCAR – <https://www.oscar-system.org/> – to which I added/modified over 160,000 lines of code), data science, and machine learning.

INTRODUCTION TO STRING THEORY

Black holes are extremely massive objects where gravitational forces dominate the surrounding region. However, near their centers, where spacetime is highly curved and the scale is incredibly small, quantum effects become significant. These interactions between quantum mechanics and gravity remain one of the major conceptual challenges in theoretical physics. Motivated by these phenomena, the physics community has long sought a unified description of the four fundamental forces: electromagnetism, the weak and strong nuclear interactions, and gravity. [1–4]. The *standard model of particle physics* provides a unified framework for the first three forces [5, 6], but gravity’s integration remains elusive [7]. *String theory* offers a potential solution, proposing that elementary particles are small vibrating strings [8–11].

A consistent quantum description of strings requires a 10-dimensional spacetime $\mathcal{S}$ [8–11], differing from our familiar 4 dimensions (time plus three spatial directions). This is reconciled through compactification, with $\mathcal{S}$ often expressed as $\mathcal{E} \times \mathcal{M}$, where $\mathcal{E}$ is the observable 4-dimensional spacetime and $\mathcal{M}$ a compact 6-dimensional Calabi-Yau manifold. Experimental endeavors have yet to yield evidence for extra spacetime dimensions, leading to the presumption that $\mathcal{M}$ is exceptionally minuscule. The geometry of $\mathcal{M}$ significantly impacts the physics observed in $\mathcal{E}$, and identifying geometries $\mathcal{M}$ to match string theory’s predictions with observations remains a critical challenge.

String theory is governed by a functional, where sections of vector bundles represent elementary particles. Minimizing this functional leads to differential equations that dictate particle dynamics. For consistency, each particle must have a “partner,” a concept known as supersymmetry. Though central to the theory, supersymmetry remains experimentally unconfirmed. It is worth recalling that the Higgs boson [12, 13] took fifty years to discover, and *gravitational waves* required a century. I remain optimistic that supersymmetry will eventually be validated, e.g. through supersymmetry breaking mechanisms [14, 15].

String phenomenology aims to reconcile string theory with our experimentally observed physics. This involves identifying suitable geometries $\mathcal{M}$ or proving their absence. Enormous efforts in this direction have been undertaken. Broadly, these research efforts diverge into two categories: matching string theory with cosmological observations and aligning it with particle accelerator results. My research focuses on the latter.

String theory can be defined by five different, yet equivalent, actions [8, 9, 16]. Early efforts in string phenomenology focused on the $E_8 \times E_8$ heterotic string [17–24] and eventually expanded to type IIA and IIB string theory [25–32]. For these solutions, one assumes that a parameter – the *interaction strength* – is small enough to ensure convergence of Taylor series, which are used to approximate solutions to the differential equations resulting from minimizing the action. These solutions are collected termed the perturbative sector of string theory. The first perturbative realization of the minimally supersymmetric standard model (MSSM) – a minimal extension of the empirically-backed standard model of particle physics via supersymmetry – exists in [20, 21] with further insights in [33, 34]. However, many other perturbative models introduce unobserved exotic particles. A prevailing challenge, as noted in [22–24] and detailed in [35], is that some observed particle interactions are omitted or differ significantly from experiments like those at CERN.

Alongside ongoing studies on (perturbative) heterotic line bundle standard models [22–24, 35], significant attention has been given to *F-theory* in probing strongly coupled IIB string theory. F-theory adeptly bridges the gap between geometry and physics [36–38], enriched by techniques from algebraic geometry. Due to its geometric consistency, F-theory supports all particle interactions needed for consistency. Pioneering studies in this realm encompass references such as [39–51]. A milestone in this domain is the discovery of the *Quadrillion F-theory standard models (QSMs)* [52], renowned for their physically appealing properties (global consistency, gauge coupling unification, and absence of chiral exotics). The QSMs stand out as the largest known class (more than 10^{15} geometries) of F-theory standard model solutions with these attributes.

MY PAST CONTRIBUTIONS

Higgs pairs are a vital component in our understanding of particle physics, as underpinned by the Nobel Prize awarded to Higgs and Englert [12, 13]. For instance, a *single* Higgs pair is imperative for the MSSM. To check for alignment among F-theory solutions and the MSSM, it is thus imprudent to compute the total number of Higgs pairs. This leads to the study of vector-like spectra. Those spectra are encoded in line bundle cohomology on smooth, irreducible curves of high genus, which among others introduces challenges due to the curve's continuous Picard group. My engagement with vector-like spectra in F-theory is not only rooted in the dire need to investigate this property of F-theory solutions, but also inspired by the rich mathematical tapestry of this topic, which invites the application of modern mathematics and computational tools. My explorations encompass cohomologies of coherent sheaves on toric varieties [44, 48, 50], Freyd categories [53], and machine learning enhancements [54]. More recently, I focused on the F-theory QSMs [55–59], for which cohomologies of root bundles encode the vector-like spectra. Significant progress towards the Brill-Noether theory of root bundles on nodal curves has been made, which serves as upper bounds for the vector-like spectra of the QSMs.

Navigating the computational challenges in F-theory can be arduous. This slows progress, limits the exploration of complex geometries, and presents a steep entry barrier for newcomers. In collaboration with my colleague A. Turner (Virginia Tech, USA), I have initiated a program to enrich the OSCAR computer algebra system [60, 61] with tools specifically designed for F-theory applications [62]. As of now, I have added and modified more than 160,000 lines of code, with significant contributions to its toric geometry functionality [63, 64]. OSCAR uses the modern Julia programming language, a significant advantage over earlier attempts [65–67]. A notable feature of the **FTheoryTools** is its capability to effortlessly extract and modify geometric constructions from existing literature. Another key feature are the advanced singularity resolution techniques – F-theory is inherently defined on a singular space, but properties of the underlying physics are easiest extracted from a resolution of this singular space. Advancing these developments and establishing **FTheoryTools** as a recognized computational tool in the F-theory community is one of my main future research goals.

VISION FOR PRACTICAL ADVANCES: FTHEORYTOOLS

To fully appreciate the computational challenges in F-theory, a brief introduction in order. Recall that type IIB string theory, just as the other five perturbative string theories, are defined by a functional, which governs functions – or rather sections of vector bundles – that represent elementary particles. Among those functions – often termed *fields* in physics lingo – the dilaton field ϕ and the RR gauge potentials C_0 , C_8 are of ample importance. The dilaton field's significance stems from its linkage to the strength of string interactions [8–11]. Interestingly, C_8 's equations of motion (the differential equation for C_8 resulting from minimizing the type IIB action) rule out any non-trivial solutions on a smooth space $\mathcal{M}$. The application of an involution σ to $\mathcal{M}$, i.e. $\mathcal{M} \rightarrow \mathcal{B} := \mathcal{M}/\sigma$, leads to a space $\mathcal{B}$ which allows for non-trivial solutions to the C_8 equations of motion. Spaces akin to $\mathcal{B}$ serve as the foundation for type IIB orientifold theories [14, 68, 69]. In such orientifold theories, the dilaton field ϕ can manifest singularities, implying an infinite/undefined value at specific loci, leading to strong string interactions. Such strength contradicts the essential premise of weak interactions in perturbation theory. This necessitates a transition from the perturbative type IIB supergravity description to a non-perturbative framework [70], namely *F-theory* [36]. To this end, the dilaton ϕ and RR gauge field C_0 merge into the axio-dilaton τ :

$$\tau: \mathcal{E} \times \mathcal{B} \rightarrow \mathbb{C}, \quad x \mapsto C_0(x) + ie^{-\phi(x)}.$$

Due to *Lorentz invariance*, τ is constant on $\mathcal{E}$ and a section of a holomorphic $\mathrm{SL}(2, \mathbb{Z})$ line bundle over $\mathcal{B}$ [37, 38]. For any $x \in \mathcal{B}$, we may thus understand $\tau(x)$ as the complex structure modulus of an elliptic curve and an elliptically fibered 4-fold $\pi: Y_4 \rightarrow \mathcal{B}$ with fibre $\mathbb{C}_{1, \tau(x)}$ can serve as “book-keeping” device of the axio-dilaton. The geometry of Y_4 enforces consistency in that it ensures that the equation of motion for C_8 has a solution and leads to the encoded axio-dilaton field τ . In addition, Y_4 encodes a lot of information about the underlying physics [71–73]. Non-trivial physics necessitates a singular Y_4 . In need for better alternatives, it is common to crepantly resolve Y_4 [74]. My research assumes that, up to $\mathbb{Q}$ -factorial terminal singularities, at least one crepant resolution $\hat{Y}_4$ does exist.

Goals and Features of FTheoryTools

In F-theory setups, the initial geometric challenge is the crepant resolution of Y_4 . A comprehensive algorithm is still missing, especially in determining $\mathbb{Q}$ -factorial terminal singularities. Typically, we apply the entire toolkit to singularities, presuming absence of a crepant resolution when those standard methods fail. Hence, incorporating state-of-the-art resolution routines, for instance including the weighted blowups explored in [75], is paramount. An equally significant feature is a database to automatically utilize established literature constructions, including the set of known resolutions.

Upon resolution, the ensuing step involves examining the given geometry using topological tools, notably through the application of pushforward formulae [47, 76]. This technique facilitates the translation of intersection theory computations from the resolved 4-fold $\widehat{Y}_4$ to the base $\mathcal{B}$, often simplifying the calculations and revealing patterns. For instance, it shows that specific physical quantities solely depend on a base intersection number, as seen in F-theory QSMs which hinge on the triple intersection number of the anticanonical class of the base 3-fold. Enhancing the **FTheoryTools** with intersection theory and topological intersection numbers, and venturing beyond the toric regime, presents collaborative prospects.

Incorporating prevalent F-theory methodologies into the **FTheoryTools** offers an excellent avenue for students to delve into advanced research, interact with relevant (computer) geometries, and contribute to our database. Although concise, these interactions hold significant potential for deeper exploration. For instance, similar to [54], this database may be used for machine learning and data science explorations [77]. A case in point is to probe a theory of Brill-Noether numbers or, if unattainable, to investigate a then promising cryptosystem. I will elaborate on this avenue in larger detail below.

The singularities of Y_4 determine a non-abelian (gauge) group. To make contact with the observed physical laws, it is advantageous to augment this group with abelian group factors, as this allows to enforce so-called selection rules. These abelian factors originate from the torsion-free subgroup of the Mordell–Weil group of $\widehat{Y}_4$, which represents the group of infinite-order rational sections of the fibration, governed by elliptic curve addition. Consequently, the rank of the abelian part of the gauge algebra aligns with the Mordell–Weil rank. It is noteworthy that the torsional part of the Mordell–Weil rank relates to the gauge group’s global structure in the physical theory. Similarly, exploring F-theory on elliptic fibrations with multi-sections can be pursued, the study of which leads to the Weil–Châtelet group and discrete factors thereof [78].

To explore alignment of F-theory solutions with empirical observations in particle physics, it is critical to understand the chiral spectrum [39, 41–43, 45–47, 76, 79]. It remains to implement algorithms into **FTheoryTools**, which facilitate this task, e.g. by use of the renowned `cohomCalc`-algorithm [80–84]. In extending towards vector-like spectra, which count the Higgs pairs in a given F-theory solution, foundational investigations into Deligne cohomology and root bundles are necessary.

VISION FOR THEORY ADVANCES: SPIN BUNDLES & BRILL-NOETHER NUMBERS

The F-theory QSMs [52] provide 10^{15} solutions apt for the standard model of particle physics. My investigation into their vector-like spectra – a crucial ingredient to compare these solutions to experimental findings from particle accelerators – directed me to root bundles on nodal curves. Before we explain how this topic arises from the physics, I wish to first provide a noval introduction to Brill-Noether numbers and subsequently elaborate on the significance of these numbers for F-theory.

Brill-Noether Numbers – A Noval Introduction

Root bundles generalize spin bundles on a smooth, irreducible genus g curve. The r -th root bundles ($r \in \mathbb{Z}_{\geq 2}$) of a given divisor class E correspond to those divisor classes D with $r \cdot D = E$. E is not necessarily the canonical divisor and there are r^{2g} roots, if existent. Nodal curves $C^\bullet$ bring two nuances: they typically have multiple irreducible components due to nodal singularities, and their root bundles can be enumerated as limit roots [85] (cf. [86–88]). Here is the dual graph (vertices are irreducible components and edges nodal singularities) of a nodal curve for which all irreducible components are $\mathbb{P}^1$ s:



Our objective is to enumerate limit roots $P^\bullet$ with $12P^\bullet = 12K_{C^\bullet}$ and discern their global section count. As we shall motivate below, this count of global sections filters F-theory solutions potentially aligned with experimental findings. Here, there are 12^8 limit roots, of which 12^4 roots have $h^0(C^\bullet, P^\bullet) = 4$ and the rest three global sections:

Roots Count	$h^0 = 3$	$h^0 \geq 3$	$h^0 = 4$	$h^0 \geq 4$
12^8	$12^4 \cdot (12^4 - 1)$	0	12^4	0

Although the number of global sections for some roots $P^\bullet$ cannot be uniquely determined, an optimal lower bound can be computed, as shown in every second column of the previous table for more complex examples. This leads to the partition: $12^8 = 12^4 \cdot (12^4 - 1) + 0 + 12^4 + 0$. The order of the summands is crucial, and summands can reappear. Our technology recently culminated in [58], computing in a certain sense an optimal partition, which likely bears a deeper meaning and carries a striking resemblance to classical Brill-Noether theory [89]. This leads me to dub the summands *Brill-Noether numbers*.

Spin Bundle: A Critical Piece in F-theory’s Puzzle

Vector-like spectra are essential in analyzing the matter fields in string theory solutions. Those matter fields include the Higgs pair [12, 13], a salient component highlighted by the Nobel Prize awarded to Higgs and Englert. The matter fields manifest

as section of line bundles over *matter curves* $C_{\mathbf{R}} \subset \mathcal{B}$ [90–92]. The matter curves and the group representations of the matter fields are in turn determined by Y_4 's singularities [73]. The sections, which represent the matter fields, are determined by gauge fields. The latter are described by the Deligne cohomology $H_D^4(Y_4, \mathbb{Z}(2))$ [93–99]. The results in [44] imply that these gauge fields induce a line bundle $\mathcal{L}_{\mathbf{R}}$ on $C_{\mathbf{R}}$. Classical results show that the chiral matter fields are elements of $H^0(C_{\mathbf{R}}, \mathcal{L}_{\mathbf{R}} \otimes \sqrt{K_{C_{\mathbf{R}}}})$ [100, 101].¹ On a curve of genus g , the choice of the spin bundle $\sqrt{K_{C_{\mathbf{R}}}}$ from the 2^{2g} possibilities significantly influences the number of sections of $L_{\mathbf{R}} = \mathcal{L}_{\mathbf{R}} \otimes \sqrt{K_{C_{\mathbf{R}}}}$. Identifying the spin bundle selected by the physics is essential but complex [91, 102]. Recent advances [59] underscore the necessity of addressing this question.

Brill-Noether Numbers: Upper Bound to Vector-Like Spectra of F-theory QSMs

The original work on F-theory QSMs [52] does not investigate the vector-like spectra. We deduced the pivotal constraint that $L_{\mathbf{R}}$ is a specific root bundle $P_{\mathbf{R}}$ [55]. On the quark-doublet curve $C_{(3,2)_{1/6}}$, this root bundle satisfies $P_{(3,2)_{1/6}}^{\otimes 2\bar{K}_{\mathcal{B}}^3} = K_{(3,2)_{1/6}}^{\otimes (6+\bar{K}_{\mathcal{B}}^3)}$, where $\bar{K}_{\mathcal{B}}^3 \in \{6, 10, 18, 30\}$. A limited number of all solutions to this constraint will arise from an apt choice of the spin bundle. Instead of pinpointing this correct choice, we took a statistical approach: Enumerate all solutions to the root bundle constraint, count the global sections of all roots, and determined probable F-theory geometries devoid of absence of exotic vector-like quark-doublets. Enumerating all roots and working out their global section count is close to impossible for smooth, irreducible curves of genus $g > 2$. Indeed, $2g = \bar{K}_{\mathcal{B}}^3 + 2$ for $C_{(3,2)_{1/6}}$. Brill-Noether numbers offer a practical upper bound:

1. Deform $C_{(3,2)_{1/6}}$ into $C_{(3,2)_{1/6}}^{\bullet}$, which is shared across various geometries [52] due to their origin from toric K3-surfaces desingularizations [56], cf. [103–105]. A computer scan of the majority of the 10^{15} QSM geometries ensues [52, 106].
2. Employ techniques from [85] (cf. [86–88]) to list all *limit root* $P_{(3,2)_{1/6}}^{\bullet}$ with our software [107].
3. Compute the global sections of each limit root $P_{(3,2)_{1/6}}^{\bullet}$ with the techniques developed in [55–57], which recently culminated in an optimal approach [58]. This leads to the Brill-Noether numbers introduced above.
4. If $h^0(C_{(3,2)_{1/6}}^{\bullet}, P_{(3,2)_{1/6}}^{\bullet}) > \chi(P_{(3,2)_{1/6}})$, the number of global sections may decrease as we smooth $C_{(3,2)_{1/6}}^{\bullet}$ into $C_{(3,2)_{1/6}}$. Hence, the Brill-Noether numbers serve as upper bound to the desired numbers on $C_{(3,2)_{1/6}}$.

Question 1: Towards a Theory of Brill-Noether Numbers

Even though the computation of the Brill-Noether numbers are resource-intensive, we currently use them as upper bound to the F-theory QSMs' vector-like spectra. A deeper understanding of the link between a nodal curve and these numbers is desired. One may posit if the Brill-Noether numbers can be inferred from the nodal curve's dual graph and the root bundle constraint. Employing both machine learning tools and analytic/algebraic insights can be beneficial in this pursuit, mimicking efforts in [54]. If the systematics are revealed, it opens doors for analogous analyses on the intricate Higgs curve of the F-theory QSMs. To illustrate the Higgs curve's complexity, consider a base $\mathcal{B}$ with $\bar{K}_{\mathcal{B}}^3 = 6$. Then, $g(C_{(3,2)_{1/6}}) = 4$, while $g(C_{(1,2)_{-1/2}}) = 28$. Not only are there many more limit roots to be enumerated, but also the dual graph is more complicated.

Question 2: An F-Theory-Inspired Cryptosystem?

For a given integer partition, can we identify a graph and a root bundle constraint, such that the ensuing Brill-Noether numbers match exactly with the initial partition? The inverse of this question is computationally extremely demanding with our methods. Attempting the direct approach seems daunting. This disparity raises the possibility of a new cryptosystem.

Question 3: Jumps Meet Yukawa Interactions

A Higgs pair [12, 13] is present in an F-theory solution exactly if

$$h^0(C_{(1,2)_{-1/2}}, P_{(1,2)_{-1/2}}) = h^1(C_{(1,2)_{-1/2}}, P_{(1,2)_{-1/2}}) = 1. \quad (3)$$

However, as $h^0(C_{(1,2)_{1/6}}^{\bullet}, P_{(1,2)_{1/6}}^{\bullet}) > \chi(P_{(1,2)_{1/6}}^{\bullet})$, the counts on $C_{(1,2)_{1/6}}^{\bullet}$ are merely upper bounds and jumping phenomena may occur, just as in classical Brill-Noether theory [89, 108]. To refine the analysis, we must understand the differences between the global section counts on $C_{(1,2)_{-1/2}}^{\bullet}$ and $C_{(1,2)_{-1/2}}$. The physics expects interactions at the nodes of $C_{(1,2)_{-1/2}}^{\bullet}$, which may be summarized in a so-called mass matrix M , whose rank is expected to be the difference in global section counts. The challenge lies in computing M [109, 110] and aligning with e.g. the mathematics of *limit linear series* [111–113].

¹The elements of $H^1(C_{\mathbf{R}}, \mathcal{L}_{\mathbf{R}} \otimes \sqrt{K_{C_{\mathbf{R}}}})$ are anti-chiral matter fields. "Anti-chiral" means that they transform in the charge conjugate rep. $\bar{\mathbf{R}}$.

REFERENCES

- [1] Glashow, S. L., *Partial-symmetries of weak interactions*, Nuclear Physics **22**, no. 4, 1961, [https://doi.org/10.1016/0029-5582\(61\)90469-2](https://doi.org/10.1016/0029-5582(61)90469-2).
- [2] Weinberg, S., *A Model of Leptons*, Phys. Rev. Lett. **19**, Nov 1967, <https://doi.org/10.1103/PhysRevLett.19.1264>.
- [3] Salam, A., *Weak and electromagnetic interactions*, Selected Papers of Abdus Salam pp. 244–254, 1994, https://doi.org/10.1142/9789812795915_0034.
- [4] Einstein, A., *Die Feldgleichungen der Gravitation*, pp. 88–92, John Wiley & Sons, Ltd, 2005, ISBN 9783527608959, <https://doi.org/10.1002/3527608958.ch5>.
- [5] Griffiths, D., *Introduction to Elementary Particles*, Physics textbook, Wiley, 2008, ISBN 9783527618477, <https://doi.org/10.1002/9783527618460>.
- [6] Schwartz, M. D., *Quantum Field Theory and the Standard Model*, Cambridge University Press, 2014, ISBN 9781107034730, <https://doi.org/10.1017/9781139540940>.
- [7] Shomer, Assaf, *A Pedagogical explanation for the non-renormalizability of gravity* 2007, [[0709.3555](https://arxiv.org/abs/0709.3555)].
- [8] Polchinski, J., *String Theory: Volume 1, An Introduction To The Bosonic String*, Cambridge monographs on mathematical physics, Cambridge University Press, 1998, ISBN 9781139457408, <https://doi.org/10.1017/CB09780511816079>.
- [9] Polchinski, J., *String Theory: Volume 2, Superstring Theory and Beyond*, Cambridge Monographs on Mathematical Physics, Cambridge University Press, 2005, ISBN 9780521672283, <https://doi.org/10.1017/CB09780511618123>.
- [10] Green, M. B. and Schwarz, J. H. and Witten, E., *Superstring Theory: Volume 1, Introduction*, Cambridge Monographs on Mathematical Physics, Cambridge University Press, 1988, ISBN 9780521357524, <https://doi.org/10.1017/CB09781139248563>.
- [11] Green, M. B. and Schwarz, J. H. and Witten, E., *Superstring Theory: Volume 2, Loop Amplitudes, Anomalies and Phenomenology*, Cambridge Monographs on Mathematical Physics, Cambridge University Press, 1987, ISBN 9780521357531, <https://doi.org/10.1017/CB09781139248570>.
- [12] Englert, F. and Brout, R., *Broken Symmetry and the Mass of Gauge Vector Mesons*, Phys. Rev. Lett. **13**, Aug 1964, <https://doi.org/10.1103/PhysRevLett.13.321>.
- [13] Higgs, P. W., *Broken Symmetries and the Masses of Gauge Bosons*, Phys. Rev. Lett. **13**, Oct 1964, <https://doi.org/10.1103/PhysRevLett.13.508>.
- [14] Blumenhagen, R. and Körs, B. and Lüst, D. and Stieberger, S., *Four-dimensional string compactifications with D-branes, orientifolds and fluxes*, Physics Reports **445**, Jul 2007, <https://doi.org/10.1016/j.physrep.2007.04.003>.
- [15] Ibáñez, L. E. and Uranga, A. M., *String Theory and Particle Physics: An Introduction to String Phenomenology*, Cambridge University Press, 2012, ISBN 9780521517522, <https://doi.org/10.1017/CB09781139018951>.
- [16] Blumenhagen, R. and Lüst, D. and Theisen, S., *Basic Concepts of String Theory*, Theoretical and Mathematical Physics, Springer, 2012, ISBN 9783642294969, <https://doi.org/10.1007/978-3-642-29497-6>.
- [17] Candelas, P. and Horowitz, G. T. and Strominger, A. and Witten, E., *Vacuum configurations for superstrings*, Nuclear Physics B **258**, 1985, [https://doi.org/10.1016/0550-3213\(85\)90602-9](https://doi.org/10.1016/0550-3213(85)90602-9).
- [18] Greene, B. R. and Kirklín, K. H. and Miron, P. J. and Ross, G. G., *A superstring-inspired standard model*, Physics Letters B **180**, no. 1, 1986, [https://doi.org/10.1016/0370-2693\(86\)90137-1](https://doi.org/10.1016/0370-2693(86)90137-1).
- [19] Braun, V. and He, Y.-H. and Ovrut, B. A. and Pantev, T., *A heterotic standard model*, Physics Letters B **618**, no. 1-4, Jul 2005, <https://doi.org/10.1016/j.physletb.2005.05.007>.

- [20] Bouchard, V. and Donagi, R., *An $SU(5)$ heterotic standard model*, Physics Letters B **633**, no. 6, Feb 2006, <https://doi.org/10.1016/j.physletb.2005.12.042>.
- [21] Bouchard, V. and Cvetič, M. and Donagi, R., *Tri-linear couplings in an heterotic minimal supersymmetric Standard Model*, Nuclear Physics B **745**, no. 1-2, Jun 2006, <https://doi.org/10.1016/j.nuclphysb.2006.03.032>.
- [22] Anderson, L. B. and Gray, J. and He, Y.-H. and Lukas, A., *Exploring positive monad bundles and a new heterotic standard model*, Journal of High Energy Physics **2010**, no. 2, Feb 2010, [https://doi.org/10.1007/JHEP02\(2010\)054](https://doi.org/10.1007/JHEP02(2010)054).
- [23] *Two hundred heterotic standard models on smooth Calabi-Yau threefolds*, Physical Review D **84**, no. 10, Nov 2011, <https://doi.org/10.1103/PhysRevD.84.106005>.
- [24] Anderson, L. B. and Gray, J. and Lukas, A. and Palti, E., *Heterotic line bundle standard models*, Journal of High Energy Physics **2012**, no. 6, Jun 2012, [https://doi.org/10.1007/JHEP06\(2012\)113](https://doi.org/10.1007/JHEP06(2012)113).
- [25] Berkooz, M. and Douglas, M. R. and Leigh, R. G., *Branes intersecting at angles*, Nuclear Physics B **480**, Nov 1996, [https://doi.org/10.1016/S0550-3213\(96\)00452-X](https://doi.org/10.1016/S0550-3213(96)00452-X).
- [26] Aldazabal, G. and Franco, S. and Ibáñez, L. E. and Rabadan, R. and Uranga, A. M., *$D=4$ chiral string compactifications from intersecting branes*, Journal of Mathematical Physics **42**, no. 7, Jul 2001, <https://doi.org/10.1063/1.1376157>.
- [27] Aldazabal, G. and Franco, S. and Ibáñez, L. E. and Rabadan, R. and Uranga, A. M., *Intersecting brane worlds*, Journal of High Energy Physics **2001**, no. 2, Feb 2001, <https://doi.org/10.1088/1126-6708/2001/02/047>.
- [28] Ibáñez, L. E. and Marchesano, F. and Rabadan, R., *Getting just the standard model at intersecting branes*, Journal of High Energy Physics **2001**, no. 11, Nov 2001, <https://doi.org/10.1088/1126-6708/2001/11/002>.
- [29] Blumenhagen, R. and Kors, B. and Lüst, D. and Ott, T., *The Standard Model from stable intersecting brane world orbifolds*, Nuclear Physics B **616**, Nov 2001, [https://doi.org/10.1016/S0550-3213\(01\)00423-0](https://doi.org/10.1016/S0550-3213(01)00423-0).
- [30] Cvetič, M. and Shiu, G. and Uranga, A. M., *Three-Family Supersymmetric Standardlike Models from Intersecting Brane Worlds*, Physical Review Letters **87**, no. 20, Oct 2001, <https://doi.org/10.1103/PhysRevLett.87.201801>.
- [31] Cvetič, M. and Shiu, G. and Uranga, A. M., *Chiral four-dimensional $N=1$ supersymmetric type IIA orientifolds from intersecting $D6$ -branes*, Nuclear Physics B **615**, Nov 2001, [https://doi.org/10.1016/S0550-3213\(01\)00427-8](https://doi.org/10.1016/S0550-3213(01)00427-8).
- [32] Blumenhagen, R. and Cvetič, M. and Langacker, P. and Shiu, G., *Toward realistic intersecting D-brane models*, Annual Review of Nuclear and Particle Science **55**, no. 1, Dec 2005, <https://doi.org/10.1146/annurev.nucl.55.090704.151541>.
- [33] Gómez, T. L. and Lukic, S. and Sols, I., *Constraining the Kähler Moduli in the Heterotic Standard Model*, Communications in Mathematical Physics **276**, no. 1, Sep 2007, <https://doi.org/10.1007/s00220-007-0338-8>.
- [34] Bouchard, V. and Donagi, R., *On heterotic model constraints*, Journal of High Energy Physics, no. 8, Aug 2008, <https://doi.org/10.1088/1126-6708/2008/08/060>.
- [35] Brodie, C. and Constantin, A. and Gray, J. and Lukas, A. and Ruehle, F., *Recent Developments in Line Bundle Cohomology and Applications to String Phenomenology*, in *Nankai Symposium on Mathematical Dialogues: In celebration of S.S.Chern's 110th anniversary*, Dec 2021, [2112.12107].
- [36] Vafa, C., *Evidence for F-theory*, Nuclear Physics B **469**, Jun 1996, [https://doi.org/10.1016/0550-3213\(96\)00172-1](https://doi.org/10.1016/0550-3213(96)00172-1).
- [37] Morrison, D. R. and Vafa, C., *Compactifications of F-theory on Calabi-Yau threefolds (II)*, Nuclear Physics B **476**, Sep 1996, [https://doi.org/10.1016/0550-3213\(96\)00369-0](https://doi.org/10.1016/0550-3213(96)00369-0).
- [38] Morrison, D. R. and Vafa, C., *Compactifications of F-theory on Calabi-Yau threefolds. (I)*, Nuclear Physics B **473**, Aug 1996, [https://doi.org/10.1016/0550-3213\(96\)00242-8](https://doi.org/10.1016/0550-3213(96)00242-8).

- [39] Grimm, T. W. and Hayashi, H., *F-theory fluxes, chirality and Chern-Simons theories*, Journal of High Energy Physics **2012**, no. 3, Mar 2012, [https://doi.org/10.1007/JHEP03\(2012\)027](https://doi.org/10.1007/JHEP03(2012)027).
- [40] Krause, S. and Mayrhofer, C. and Weigand, T., *G_4 -flux, chiral matter and singularity resolution in F-theory compactifications*, Nuclear Physics B **858**, no. 1, May 2012, <https://doi.org/10.1016/j.nuclphysb.2011.12.013>.
- [41] Krause, S. and Mayrhofer, C. and Weigand, T., *Gauge Fluxes in F-theory and Type IIB Orientifolds*, Journal of High Energy Physics **2012**, no. 8, Aug 2012, [https://doi.org/10.1007/JHEP08\(2012\)119](https://doi.org/10.1007/JHEP08(2012)119).
- [42] Braun, V. and Grimm, T. W. and Keitel, J., *Geometric Engineering in Toric F-Theory and GUTs with $U(1)$ Gauge Factors*, Journal of High Energy Physics **2013**, no. 12, Dec 2013, [https://doi.org/10.1007/JHEP12\(2013\)069](https://doi.org/10.1007/JHEP12(2013)069).
- [43] Cvetič, M. and Grassi, A. and Klevers, D. and Piragua, H., *Chiral four-dimensional F-theory compactifications with $SU(5)$ and multiple $U(1)$ -factors*, Journal of High Energy Physics **2014**, no. 4, Apr 2014, [https://doi.org/10.1007/JHEP04\(2014\)010](https://doi.org/10.1007/JHEP04(2014)010).
- [44] Bies, M. and Mayrhofer, C. and Pehle, C. and Weigand, T., *Chow groups, Deligne cohomology and massless matter in F-theory* Feb 2014, [[1402.5144](https://arxiv.org/abs/1402.5144)].
- [45] Cvetič, M. and Klevers, D. and Peña, D. K. M. and Oehlmann, P.-K. and Reuter, J., *Three-family particle physics models from global F-theory compactifications*, Journal of High Energy Physics **2015**, no. 8, Aug 2015, [https://doi.org/10.1007/JHEP08\(2015\)087](https://doi.org/10.1007/JHEP08(2015)087).
- [46] Lin, L. and Mayrhofer, C. and Till, O. and Weigand, T., *Fluxes in F-theory Compactifications on Genus-One Fibrations*, Journal of High Energy Physics **2016**, no. 1, Jan 2016, [https://doi.org/10.1007/JHEP01\(2016\)098](https://doi.org/10.1007/JHEP01(2016)098).
- [47] Lin, L. and Weigand, T., *G_4 -flux and standard model vacua in F-theory*, Nuclear Physics B **913**, Dec 2016, <https://doi.org/10.1016/j.nuclphysb.2016.09.008>.
- [48] Bies, M. and Mayrhofer, C. and Weigand, T., *Gauge backgrounds and zero-mode counting in F-theory*, Journal of High Energy Physics **2017**, no. 11, Nov 2017, [https://doi.org/10.1007/JHEP11\(2017\)081](https://doi.org/10.1007/JHEP11(2017)081).
- [49] Bies, M. and Mayrhofer, C. and Weigand, T., *Algebraic Cycles and Local Anomalies in F-Theory*, Journal of High Energy Physics **2017**, no. 11, Nov 2017, [https://doi.org/10.1007/JHEP11\(2017\)100](https://doi.org/10.1007/JHEP11(2017)100).
- [50] Bies, M., *Cohomologies of coherent sheaves and massless spectra in F-theory*, Ph.D. thesis, Heidelberg U., Feb 2018, <https://doi.org/10.11588/heidok.00024045>.
- [51] Cvetič, M. and Lin, L. and Liu, M. and Oehlmann, P.-K., *An F-theory Realization of the Chiral MSSM with $\mathbb{Z}_2$ -Parity*, Journal of High Energy Physics, no. 9, Sep 2018, [https://doi.org/10.1007/JHEP09\(2018\)089](https://doi.org/10.1007/JHEP09(2018)089).
- [52] Cvetič, M. and Halverson, J. and Lin, L. and Liu, M. and Tian, J., *Quadrillion F-Theory Compactifications with the Exact Chiral Spectrum of the Standard Model*, Physical Review Letters **123**, no. 10, Sep 2019, <https://doi.org/10.1103/PhysRevLett.123.101601>.
- [53] Bies, M. and Posur, S., *Tensor products of finitely presented functors*, Journal of Algebra and Its Applications Jul 2021, <https://doi.org/10.1142/s0219498822501869>.
- [54] Bies, M. and Cvetič, M. and Donagi, R. and Lin, L. and Liu, M. and Ruehle, F., *Machine Learning and Algebraic Approaches towards Complete Matter Spectra in 4d F-theory*, Journal of High Energy Physics **2021**, no. 1, Jan 2021, [https://doi.org/10.1007/JHEP01\(2021\)196](https://doi.org/10.1007/JHEP01(2021)196).
- [55] Bies, M. and Cvetič, M. and Donagi, R. and Liu, M. and Ong, M., *Root bundles and towards exact matter spectra of F-theory MSSMs*, Journal of High Energy Physics **2021**, no. 9, Sep 2021, [https://doi.org/10.1007/JHEP09\(2021\)076](https://doi.org/10.1007/JHEP09(2021)076).
- [56] Bies, M. and Cvetič, M. and Liu, M., *Statistics of limit root bundles relevant for exact matter spectra of F-theory MSSMs*, Physical Review D **104**, no. 6, Sep 2021, <https://doi.org/10.1103/PhysRevD.104.L061903>.
- [57] Bies, M. and Cvetič, M. and Donagi, R. and Ong, M., *Brill-Noether-general limit root bundles: absence of vector-like exotics in F-theory Standard Models*, Journal Of High Energy Physics **11**, Apr 2022, [https://doi.org/10.1007/JHEP11\(2022\)004](https://doi.org/10.1007/JHEP11(2022)004).

- [58] Bies, M. and Cvetič, M. and Donagi, R. and Ong, M., *Improved statistics for F-theory standard models*, Communications in Mathematical Physics (accepted – to appear) 2023, [[2307.02535](#)].
- [59] Bies, M., *Root bundles: Applications to F-theory Standard Models*, in Donagi, R. and Langer, A. and Sułkowski, P. and Wendland, K., editor, *String-Math 2022*, volume 107 of Proceedings of Symposia in Pure Mathematics, pp. 17–43, American Mathematical Society, 2024, ISBN 978-1-4704-7240-5, <https://bookstore.ams.org/PSPUM/107>.
- [60] The OSCAR Team, *OSCAR – Open Source Computer Algebra Research system, Version 1.0.0*, <https://www.oscar-system.org>, 2024.
- [61] Decker, W. and Eder, C. and Fieker, C. and Horn, M. and Joswig, M., editor, *The Computer Algebra System OSCAR: Algorithms and Examples*, volume 32 of Algorithms and Computation in Mathematics, Springer, 1 edition, 2024, <https://link.springer.com/book/9783031621260>.
- [62] Bies, M. and Turner, A. P., *F-Theory Applications*, volume 32 of *Algorithms and Computation in Mathematics*, Springer, 1 edition, 2024, <https://link.springer.com/book/9783031621260>.
- [63] M. Bies and L. Kastner, *Toric Geometry in OSCAR*, ComputerAlgebraRundbrief **72**, no. 2, Mar 2023, <https://fachgruppe-computeralgebra.de/data/CA-Rundbrief/car72.pdf>.
- [64] Bies, M. and Kastner, L., *Toric Geometry*, volume 32 of *Algorithms and Computation in Mathematics*, Springer, 1 edition, 2024, <https://link.springer.com/book/9783031621260>.
- [65] The Toric Varieties Project authors, *The Toric Varieties Project*, https://github.com/homalg-project/ToricVarieties_project, 2019–2024.
- [66] The homalg project authors, *The homalg project – Algorithmic Homological Algebra*, <https://homalg-project.github.io/>, 2003–2024.
- [67] Gutsche, S. and Skartsæterhagen, Ø. and Posur, S., *The CAP project – Categories, Algorithms, Programming*, http://homalg-project.github.io/CAP_project, 2013–2024.
- [68] Marchesano, F., *Intersecting D-brane models*, Ph.D. thesis, Universidad Autónoma De Madrid, May 2003, [[hep-th/0307252](#)].
- [69] Cremades, D. and Ibáñez, L. E. and Marchesano, F., *Intersecting Brane Models of Particle Physics and the Higgs Mechanism*, Journal of High Energy Physics **2002**, no. 7, Jul 2002, <https://doi.org/10.1088/1126-6708/2002/07/022>.
- [70] Greene, B. R. and Shapere, A. and Vafa, C. and Yau, S.-T., *Stringy cosmic strings and noncompact Calabi-Yau manifolds*, Nuclear Physics B **337**, no. 1, 1990, [https://doi.org/10.1016/0550-3213\(90\)90248-C](https://doi.org/10.1016/0550-3213(90)90248-C).
- [71] Denef, F., *Les Houches Lectures on Constructing String Vacua*, Les Houches **87**, 2008, [[0803.1194](#)].
- [72] Cvetič, M. and Lin, L., *TASI Lectures on Abelian and Discrete Symmetries in F-theory*, PoS **TASI2017**, 2018, <https://doi.org/10.22323/1.305.0020>.
- [73] Weigand, T., *F-theory*, PoS **TASI2017**, 2018, <https://doi.org/10.22323/1.305.0016>.
- [74] Griffiths, P. and Harris, J., *Principles of Algebraic Geometry*, Wiley Classics Library, Wiley, 2011, ISBN 9781118030776, <https://doi.org/10.1002/9781118032527>.
- [75] Arena, V. and Jefferson, P. and Obinna, S., *Intersection Theory on Weighted Blowups of F-theory Vacua* Apr 2023, [[2305.00297](#)].
- [76] Jefferson, P. and Turner, A. P., *Generating functions for intersection products of divisors in resolved F-theory models*, Nucl. Phys. B **991**, 2023, <https://doi.org/10.1016/j.nuclphysb.2023.116177>.
- [77] Ruehle, F., *Data science applications to string theory*, Physics Reports **839**, Jan 2020, <https://doi.org/10.1016/j.physrep.2019.09.005>.

- [78] Morrison, D. R. and Taylor, W., *Sections, multisections, and $U(1)$ fields in F -theory*, Journal of Singularities **15**, Oct 2016, <https://doi.org/10.5427/jsing.2016.15g>.
- [79] Jefferson, T. and Taylor, W. and Turner, A. P., *Chiral matter multiplicities and resolution-independent structure in 4D F -theory models*, Communications in Mathematical Physics **404**, Nov 2023, <https://doi.org/10.1007/s00220-023-04860-0>.
- [80] Blumenhagen, R. and Jurke, B. and Rahn, T. and Roschy, H., *Cohomology of line bundles: A computational algorithm*, Journal of Mathematical Physics **51**, no. 10, Oct 2010, <https://doi.org/10.1063/1.3501132>.
- [81] *cohomCalc package*, <http://wwwth.mppmu.mpg.de/members/blumenha/cohomcalc/>, 2010, High-performance line bundle cohomology computation based on [80].
- [82] Jow, S.-Y., *Cohomology of toric line bundles via simplicial Alexander duality*, Journal of Mathematical Physics **52**, no. 3, Mar 2011, <https://doi.org/10.1063/1.3562523>.
- [83] Roschy, H. and Rahn, T., *Cohomology of line bundles: Proof of the algorithm*, Journal of Mathematical Physics **51**, no. 10, Oct 2010, <https://doi.org/10.1063/1.3501135>.
- [84] Blumenhagen, R. and Jurke, B. and Rahn, T. and Roschy, H., *Cohomology of line bundles: Applications*, Journal of Mathematical Physics **53**, no. 1, Jan 2012, <https://doi.org/10.1063/1.3677646>.
- [85] Caporaso, L. and Casagrande, C. and Cornalba, M., *Moduli of Roots of Line Bundles on Curves*, Transactions of the American Mathematical Society **359**, no. 8, 2007, <https://doi.org/10.1090/S0002-9947-07-04087-1>.
- [86] Jarvis, T. J., *Geometry of the moduli of higher spin curves*, International Journal of Mathematics **11**, Sep 1998, <https://doi.org/10.1142/S0129167X00000325>.
- [87] Jarvis, T. J., *The Picard group of the moduli of higher spin curves*, The New York Journal of Mathematics **7**, 2001, <https://nyjm.albany.edu/j/2001/7-3.html>.
- [88] Natanzon, S. and Pratussevitch, A., *Higher Spin Klein Surfaces*, Moscow Mathematical Journal **16**, 2016, <http://www.mathjournals.org/mmj/2016-016-001/2016-016-001-004.html>.
- [89] Brill, A. and Noether, M., *Ueber die algebraischen Functionen und ihre Anwendung in der Geometrie*, Mathematische Annalen **7**, no. 2, 1874, <https://doi.org/10.1007/BF02104804>.
- [90] Katz, S. H. and Vafa, C., *Matter from geometry*, Nuclear Physics B **497**, Jul 1997, [https://doi.org/10.1016/S0550-3213\(97\)00280-0](https://doi.org/10.1016/S0550-3213(97)00280-0).
- [91] Beasley, C. and Heckman, J. J. and Vafa, C., *GUTs and exceptional branes in F -theory — I*, Journal of High Energy Physics, no. 1, Jan 2009, <https://doi.org/10.1088/1126-6708/2009/01/058>.
- [92] Donagi, R. and Wijnholt, M., *Gluing Branes — I*, Journal of High Energy Physics **2013**, no. 5, May 2013, [https://doi.org/10.1007/JHEP05\(2013\)068](https://doi.org/10.1007/JHEP05(2013)068).
- [93] Donagi, R., *Heterotic / F theory duality: ICMP lecture*, in *12th International Congress of Mathematical Physics (ICMP 97)*, pp. 206–213, Feb 1998, [[hep-th/9802093](https://arxiv.org/abs/hep-th/9802093)].
- [94] Curio, G. and Donagi, R., *Moduli in $N = 1$ heterotic/ F -theory duality*, Nuclear Physics B **518**, no. 3, May 1998, [https://doi.org/10.1016/S0550-3213\(98\)00185-0](https://doi.org/10.1016/S0550-3213(98)00185-0).
- [95] Diaconescu, E. and Freed, D. S. and Moore, G., *The M -theory 3-form and E_8 gauge theory*, pp. 44–88, London Mathematical Society Lecture Note Series, Cambridge University Press, 2007, <https://doi.org/10.1017/CB09780511721489.005>.
- [96] Freed, D. S. and Moore, G. W., *Setting the Quantum Integrand of M -Theory*, Communications in Mathematical Physics **263**, no. 1, Jan 2006, <https://doi.org/10.1007/s00220-005-1482-7>.
- [97] Intriligator, K. and Jockers, H. and Mayr, P. and Morrison, D. R. and Plesser, M. R., *Conifold transitions in M -theory on Calabi-Yau fourfolds with background fluxes*, Advances in Theoretical and Mathematical Physics **17**, no. 3, 2013, <https://doi.org/10.4310/ATMP.2013.v17.n3.a2>.

- [98] Eisenbud, D. and Harris, J., *3264 and All That: A Second Course in Algebraic Geometry*, Cambridge University Press, 2016, <https://doi.org/10.1017/CBO9781139062046>.
- [99] Fulton, W., *Intersection Theory*, Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge / A Series of Modern Surveys in Mathematics, Springer Berlin Heidelberg, 2013, ISBN 9783662024218, <https://doi.org/10.1007/978-1-4612-1700-8>.
- [100] Witten, E., *Topological sigma models*, Communications in Mathematical Physics **118**, no. 3, 1988, <https://doi.org/10.1007/BF01466725>.
- [101] Witten, E., *Mirror manifolds and topological field theory*, AMS/IP Stud. Adv. Math. **9**, 1991, [[hep-th/9112056](https://arxiv.org/abs/hep-th/9112056)].
- [102] Freed, D. S. and Witten, E., *Anomalies in string theory with D-branes*, Asian J. Math. **3**, 1999, <https://doi.org/10.4310/AJM.1999.v3.n4.a6>.
- [103] Batyrev, V. V., *Dual polyhedra and mirror symmetry for Calabi-Yau hypersurfaces in toric varieties*, J. Alg. Geom. **3**, 1994, [[alg-geom/9310003](https://arxiv.org/abs/alg-geom/9310003)].
- [104] Cox, D.A. and Katz, S., *Mirror Symmetry and Algebraic Geometry*, Mathematical surveys and monographs, American Mathematical Society, 1999, ISBN 9780821821275, <https://doi.org/https://doi.org/10.1090/surv/068>.
- [105] Kreuzer, M. and Skarke, H., *Classification of reflexive polyhedra in three-dimensions*, Adv. Theor. Math. Phys. **2**, 1998, <https://doi.org/10.4310/ATMP.1998.v2.n4.a5>.
- [106] Halverson, J. and Tian, J., *Cost of seven-brane gauge symmetry in a quadrillion F-theory compactifications*, Physical Review D **95**, no. 2, Jan 2017, <https://doi.org/10.1103/PhysRevD.95.026005>.
- [107] Bies, M., *RootCounter*, <https://github.com/Julia-meets-String-Theory/RootCounter>, 2023–2024.
- [108] Eisenbud D. and Green M. and Harris J., *Cayley-Bacharach theorems and conjectures*, Bulletin of the American Mathematical Society **33**, no. 3, Jul 1996, <https://doi.org/10.1090/s0273-0979-96-00666-0>.
- [109] Cecotti, S. and Cheng, M. C. N. and Heckman, J. J. and Vafa, C., *Yukawa Couplings in F-theory and Non-Commutative Geometry*, Surveys in differential geometry **15**, 2009, <https://doi.org/10.4310/sdg.2010.v15.n1.a3>.
- [110] Cvetič, M. and Lin, L. and Liu, M. and Zhang, H. Y. and Zoccarato, G., *Yukawa Hierarchies in Global F-theory Models*, Journal of High Energy Physics, no. 1, Jan 2020, [https://doi.org/10.1007/JHEP01\(2020\)037](https://doi.org/10.1007/JHEP01(2020)037).
- [111] Eisenbud, D. and Harris, J., *Limit linear series: Basic theory*, Inventiones mathematicae **85**, 1986, <https://doi.org/10.1007/BF01389094>.
- [112] Osserman, B., *A limit linear series moduli scheme*, Annales de l'institut Fourier **56**, no. 4, 2006, <https://doi.org/10.5802/AIF.2209>.
- [113] Farkas G., *Theta characteristics and their moduli*, Milan Journal of Mathematics **80**, 2012, <https://doi.org/10.1007/s00032-012-0178-7>.